

4.10 | Antiderivatives

Learning Objectives

- 4.10.1** Find the general antiderivative of a given function.
- 4.10.2** Explain the terms and notation used for an indefinite integral.
- 4.10.3** State the power rule for integrals.
- 4.10.4** Use antidifferentiation to solve simple initial-value problems.

At this point, we have seen how to calculate derivatives of many functions and have been introduced to a variety of their applications. We now ask a question that turns this process around: Given a function f , how do we find a function with the derivative f and why would we be interested in such a function?

We answer the first part of this question by defining antiderivatives. The antiderivative of a function f is a function with a derivative f . Why are we interested in antiderivatives? The need for antiderivatives arises in many situations, and we look at various examples throughout the remainder of the text. Here we examine one specific example that involves rectilinear motion. In our examination in **Derivatives** of rectilinear motion, we showed that given a position function $s(t)$ of an object, then its velocity function $v(t)$ is the derivative of $s(t)$ —that is, $v(t) = s'(t)$. Furthermore, the acceleration $a(t)$ is the derivative of the velocity $v(t)$ —that is, $a(t) = v'(t) = s''(t)$. Now suppose we are given an acceleration function a , but not the velocity function v or the position function s . Since $a(t) = v'(t)$, determining the velocity function requires us to find an antiderivative of the acceleration function. Then, since $v(t) = s'(t)$, determining the position function requires us to find an antiderivative of the velocity function. Rectilinear motion is just one case in which the need for antiderivatives arises. We will see many more examples throughout the remainder of the text. For now, let's look at the terminology and notation for antiderivatives, and determine the antiderivatives for several types of functions. We examine various techniques for finding antiderivatives of more complicated functions later in the text (**Introduction to Techniques of Integration** (<http://cnx.org/content/m53654/latest/>)).

The Reverse of Differentiation

At this point, we know how to find derivatives of various functions. We now ask the opposite question. Given a function f , how can we find a function with derivative f ? If we can find a function F derivative f , we call F an antiderivative of f .

Definition

A function F is an **antiderivative** of the function f if

$$F'(x) = f(x)$$

for all x in the domain of f .

Consider the function $f(x) = 2x$. Knowing the power rule of differentiation, we conclude that $F(x) = x^2$ is an antiderivative of f since $F'(x) = 2x$. Are there any other antiderivatives of f ? Yes; since the derivative of any constant C is zero, $x^2 + C$ is also an antiderivative of $2x$. Therefore, $x^2 + 5$ and $x^2 - \sqrt{2}$ are also antiderivatives. Are there any others that are not of the form $x^2 + C$ for some constant C ? The answer is no. From Corollary 2 of the Mean Value Theorem, we know that if F and G are differentiable functions such that $F'(x) = G'(x)$, then $F(x) - G(x) = C$ for some constant C . This fact leads to the following important theorem.

Theorem 4.14: General Form of an Antiderivative

Let F be an antiderivative of f over an interval I . Then,

- i. for each constant C , the function $F(x) + C$ is also an antiderivative of f over I ;
- ii. if G is an antiderivative of f over I , there is a constant C for which $G(x) = F(x) + C$ over I .

In other words, the most general form of the antiderivative of f over I is $F(x) + C$.

We use this fact and our knowledge of derivatives to find all the antiderivatives for several functions.

Example 4.50**Finding Antiderivatives**

For each of the following functions, find all antiderivatives.

- a. $f(x) = 3x^2$
- b. $f(x) = \frac{1}{x}$
- c. $f(x) = \cos x$
- d. $f(x) = e^x$

Solution

- a. Because

$$\frac{d}{dx}(x^3) = 3x^2$$

then $F(x) = x^3$ is an antiderivative of $3x^2$. Therefore, every antiderivative of $3x^2$ is of the form $x^3 + C$ for some constant C , and every function of the form $x^3 + C$ is an antiderivative of $3x^2$.

- b. Let $f(x) = \ln|x|$. For $x > 0$, $f(x) = \ln(x)$ and

$$\frac{d}{dx}(\ln x) = \frac{1}{x}.$$

For $x < 0$, $f(x) = \ln(-x)$ and

$$\frac{d}{dx}(\ln(-x)) = -\frac{1}{-x} = \frac{1}{x}.$$

Therefore,

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}.$$

Thus, $F(x) = \ln|x|$ is an antiderivative of $\frac{1}{x}$. Therefore, every antiderivative of $\frac{1}{x}$ is of the form $\ln|x| + C$ for some constant C and every function of the form $\ln|x| + C$ is an antiderivative of $\frac{1}{x}$.

- c. We have

$$\frac{d}{dx}(\sin x) = \cos x,$$

so $F(x) = \sin x$ is an antiderivative of $\cos x$. Therefore, every antiderivative of $\cos x$ is of the form $\sin x + C$ for some constant C and every function of the form $\sin x + C$ is an antiderivative of $\cos x$.

d. Since

$$\frac{d}{dx}(e^x) = e^x,$$

then $F(x) = e^x$ is an antiderivative of e^x . Therefore, every antiderivative of e^x is of the form $e^x + C$ for some constant C and every function of the form $e^x + C$ is an antiderivative of e^x .



4.49 Find all antiderivatives of $f(x) = \sin x$.

Indefinite Integrals

We now look at the formal notation used to represent antiderivatives and examine some of their properties. These properties allow us to find antiderivatives of more complicated functions. Given a function f , we use the notation $f'(x)$ or $\frac{df}{dx}$ to denote the derivative of f . Here we introduce notation for antiderivatives. If F is an antiderivative of f , we say that $F(x) + C$ is the most general antiderivative of f and write

$$\int f(x)dx = F(x) + C.$$

The symbol $\int$ is called an *integral sign*, and $\int f(x)dx$ is called the *indefinite integral* of f .

Definition

Given a function f , the **indefinite integral** of f , denoted

$$\int f(x)dx,$$

is the most general antiderivative of f . If F is an antiderivative of f , then

$$\int f(x)dx = F(x) + C.$$

The expression $f(x)$ is called the *integrand* and the variable x is the *variable of integration*.

Given the terminology introduced in this definition, the act of finding the antiderivatives of a function f is usually referred to as *integrating* f .

For a function f and an antiderivative F , the functions $F(x) + C$, where C is any real number, is often referred to as *the family of antiderivatives of f* . For example, since x^2 is an antiderivative of $2x$ and any antiderivative of $2x$ is of the form $x^2 + C$, we write

$$\int 2x \, dx = x^2 + C.$$

The collection of all functions of the form $x^2 + C$, where C is any real number, is known as the *family of antiderivatives of $2x$* . **Figure 4.85** shows a graph of this family of antiderivatives.

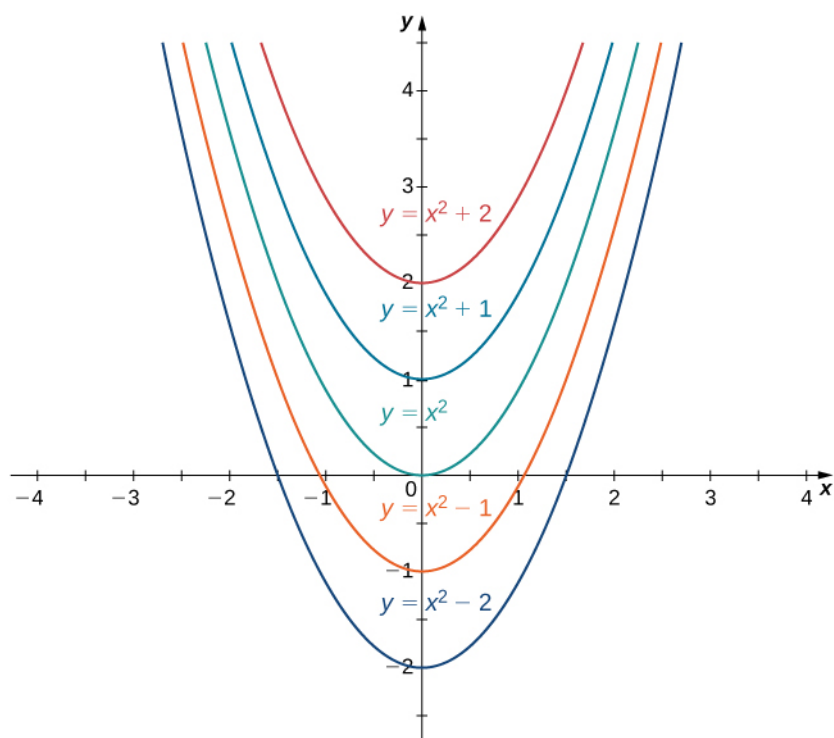


Figure 4.85 The family of antiderivatives of $2x$ consists of all functions of the form $x^2 + C$, where C is any real number.

For some functions, evaluating indefinite integrals follows directly from properties of derivatives. For example, for $n \neq -1$,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C,$$

which comes directly from

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = (n+1) \frac{x^n}{n+1} = x^n.$$

This fact is known as *the power rule for integrals*.

Theorem 4.15: Power Rule for Integrals

For $n \neq -1$,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C.$$

Evaluating indefinite integrals for some other functions is also a straightforward calculation. The following table lists the indefinite integrals for several common functions. A more complete list appears in **Appendix B**.

Differentiation Formula	Indefinite Integral
$\frac{d}{dx}(k) = 0$	$\int k dx = \int kx^0 dx = kx + C$
$\frac{d}{dx}(x^n) = nx^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + C$
$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}(\cos x) = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
$\frac{d}{dx}(\csc x) = -\csc x \cot x$	$\int \csc x \cot x dx = -\csc x + C$
$\frac{d}{dx}(\sec x) = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
$\frac{d}{dx}(\cot x) = -\csc^2 x$	$\int \csc^2 x dx = -\cot x + C$
$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} = \sin^{-1} x + C$
$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$
$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + C$

Table 4.13 Integration Formulas

From the definition of indefinite integral of f , we know

$$\int f(x) dx = F(x) + C$$

if and only if F is an antiderivative of f . Therefore, when claiming that

$$\int f(x)dx = F(x) + C$$

it is important to check whether this statement is correct by verifying that $F'(x) = f(x)$.

Example 4.51

Verifying an Indefinite Integral

Each of the following statements is of the form $\int f(x)dx = F(x) + C$. Verify that each statement is correct by showing that $F'(x) = f(x)$.

a. $\int (x + e^x)dx = \frac{x^2}{2} + e^x + C$

b. $\int xe^x dx = xe^x - e^x + C$

Solution

a. Since

$$\frac{d}{dx}\left(\frac{x^2}{2} + e^x + C\right) = x + e^x,$$

the statement

$$\int (x + e^x)dx = \frac{x^2}{2} + e^x + C$$

is correct.

Note that we are verifying an indefinite integral for a sum. Furthermore, $\frac{x^2}{2}$ and e^x are antiderivatives of x and e^x , respectively, and the sum of the antiderivatives is an antiderivative of the sum. We discuss this fact again later in this section.

b. Using the product rule, we see that

$$\frac{d}{dx}(xe^x - e^x + C) = e^x + xe^x - e^x = xe^x.$$

Therefore, the statement

$$\int xe^x dx = xe^x - e^x + C$$

is correct.

Note that we are verifying an indefinite integral for a product. The antiderivative $xe^x - e^x$ is not a product of the antiderivatives. Furthermore, the product of antiderivatives, $x^2e^x/2$ is not an antiderivative of xe^x since

$$\frac{d}{dx}\left(\frac{x^2e^x}{2}\right) = xe^x + \frac{x^2e^x}{2} \neq xe^x.$$

In general, the product of antiderivatives is not an antiderivative of a product.



4.50 Verify that $\int x \cos x dx = x \sin x + \cos x + C$.

In **Table 4.13**, we listed the indefinite integrals for many elementary functions. Let's now turn our attention to evaluating indefinite integrals for more complicated functions. For example, consider finding an antiderivative of a sum $f + g$.

In **Example 4.51a**, we showed that an antiderivative of the sum $x + e^x$ is given by the sum $\left(\frac{x^2}{2}\right) + e^x$ —that is, an antiderivative of a sum is given by a sum of antiderivatives. This result was not specific to this example. In general, if F and G are antiderivatives of any functions f and g , respectively, then

$$\frac{d}{dx}(F(x) + G(x)) = F'(x) + G'(x) = f(x) + g(x).$$

Therefore, $F(x) + G(x)$ is an antiderivative of $f(x) + g(x)$ and we have

$$\int (f(x) + g(x)) dx = F(x) + G(x) + C.$$

Similarly,

$$\int (f(x) - g(x)) dx = F(x) - G(x) + C.$$

In addition, consider the task of finding an antiderivative of $kf(x)$, where k is any real number. Since

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}F(x) = kf'(x)$$

for any real number k , we conclude that

$$\int kf(x) dx = kF(x) + C.$$

These properties are summarized next.

Theorem 4.16: Properties of Indefinite Integrals

Let F and G be antiderivatives of f and g , respectively, and let k be any real number.

Sums and Differences

$$\int (f(x) \pm g(x)) dx = F(x) \pm G(x) + C$$

Constant Multiples

$$\int kf(x) dx = kF(x) + C$$

From this theorem, we can evaluate any integral involving a sum, difference, or constant multiple of functions with antiderivatives that are known. Evaluating integrals involving products, quotients, or compositions is more complicated (see **Example 4.51b**, for an example involving an antiderivative of a product.) We look at and address integrals involving these more complicated functions in **Introduction to Integration**. In the next example, we examine how to use this theorem to calculate the indefinite integrals of several functions.

Example 4.52

Evaluating Indefinite Integrals

Evaluate each of the following indefinite integrals:

- $\int (5x^3 - 7x^2 + 3x + 4) dx$
- $\int \frac{x^2 + 4\sqrt[3]{x}}{x} dx$
- $\int \frac{4}{1+x^2} dx$
- $\int \tan x \cos x dx$

Solution

- Using **Properties of Indefinite Integrals**, we can integrate each of the four terms in the integrand separately. We obtain

$$\int (5x^3 - 7x^2 + 3x + 4) dx = \int 5x^3 dx - \int 7x^2 dx + \int 3x dx + \int 4 dx.$$

From the second part of **Properties of Indefinite Integrals**, each coefficient can be written in front of the integral sign, which gives

$$\int 5x^3 dx - \int 7x^2 dx + \int 3x dx + \int 4 dx = 5 \int x^3 dx - 7 \int x^2 dx + 3 \int x dx + 4 \int 1 dx.$$

Using the power rule for integrals, we conclude that

$$\int (5x^3 - 7x^2 + 3x + 4) dx = \frac{5}{4}x^4 - \frac{7}{3}x^3 + \frac{3}{2}x^2 + 4x + C.$$

- Rewrite the integrand as

$$\frac{x^2 + 4\sqrt[3]{x}}{x} = \frac{x^2}{x} + \frac{4\sqrt[3]{x}}{x} = 0.$$

Then, to evaluate the integral, integrate each of these terms separately. Using the power rule, we have

$$\begin{aligned} \int \left(x + \frac{4}{x^{2/3}} \right) dx &= \int x dx + 4 \int x^{-2/3} dx \\ &= \frac{1}{2}x^2 + 4 \frac{1}{\left(\frac{-2}{3}\right) + 1} x^{(-2/3) + 1} + C \\ &= \frac{1}{2}x^2 + 12x^{1/3} + C. \end{aligned}$$

- Using **Properties of Indefinite Integrals**, write the integral as

$$4 \int \frac{1}{1+x^2} dx.$$

Then, use the fact that $\tan^{-1}(x)$ is an antiderivative of $\frac{1}{(1+x^2)}$ to conclude that

$$\int \frac{4}{1+x^2} dx = 4 \tan^{-1}(x) + C.$$

- Rewrite the integrand as

$$\tan x \cos x = \frac{\sin x}{\cos x} \cos x = \sin x.$$

Therefore,

$$\int \tan x \cos x = \int \sin x = -\cos x + C.$$



4.51 Evaluate $\int (4x^3 - 5x^2 + x - 7)dx$.

Initial-Value Problems

We look at techniques for integrating a large variety of functions involving products, quotients, and compositions later in the text. Here we turn to one common use for antiderivatives that arises often in many applications: solving differential equations.

A *differential equation* is an equation that relates an unknown function and one or more of its derivatives. The equation

$$\frac{dy}{dx} = f(x) \quad (4.9)$$

is a simple example of a differential equation. Solving this equation means finding a function y with a derivative f . Therefore, the solutions of **Equation 4.9** are the antiderivatives of f . If F is one antiderivative of f , every function of the form $y = F(x) + C$ is a solution of that differential equation. For example, the solutions of

$$\frac{dy}{dx} = 6x^2$$

are given by

$$y = \int 6x^2 dx = 2x^3 + C.$$

Sometimes we are interested in determining whether a particular solution curve passes through a certain point (x_0, y_0) —that is, $y(x_0) = y_0$. The problem of finding a function y that satisfies a differential equation

$$\frac{dy}{dx} = f(x) \quad (4.10)$$

with the additional condition

$$y(x_0) = y_0 \quad (4.11)$$

is an example of an **initial-value problem**. The condition $y(x_0) = y_0$ is known as an *initial condition*. For example, looking for a function y that satisfies the differential equation

$$\frac{dy}{dx} = 6x^2$$

and the initial condition

$$y(1) = 5$$

is an example of an initial-value problem. Since the solutions of the differential equation are $y = 2x^3 + C$, to find a function y that also satisfies the initial condition, we need to find C such that $y(1) = 2(1)^3 + C = 5$. From this equation, we see that $C = 3$, and we conclude that $y = 2x^3 + 3$ is the solution of this initial-value problem as shown in the following graph.

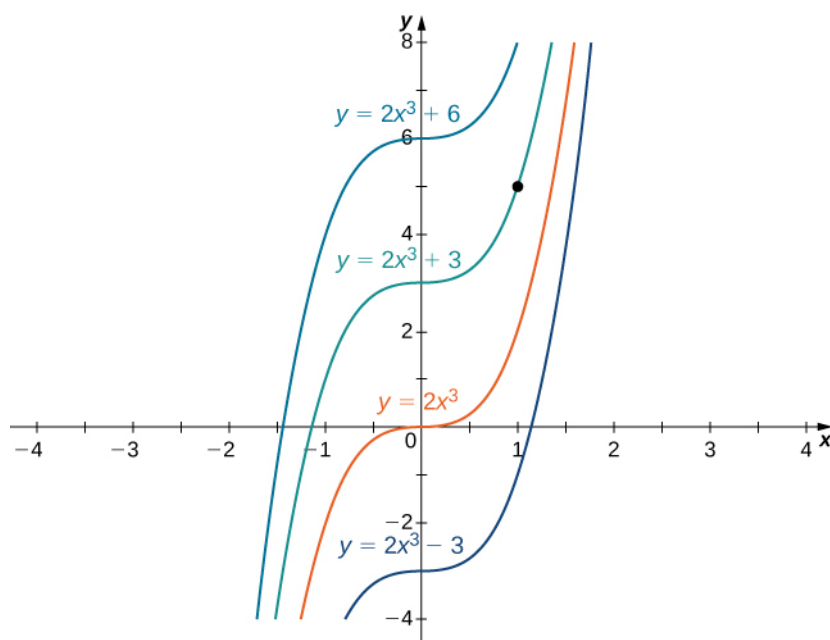


Figure 4.86 Some of the solution curves of the differential equation $\frac{dy}{dx} = 6x^2$ are displayed. The function $y = 2x^3 + 3$ satisfies the differential equation and the initial condition $y(1) = 5$.

Example 4.53

Solving an Initial-Value Problem

Solve the initial-value problem

$$\frac{dy}{dx} = \sin x, \quad y(0) = 5.$$

Solution

First we need to solve the differential equation. If $\frac{dy}{dx} = \sin x$, then

$$y = \int \sin(x) dx = -\cos x + C.$$

Next we need to look for a solution y that satisfies the initial condition. The initial condition $y(0) = 5$ means we need a constant C such that $-\cos x + C = 5$. Therefore,

$$C = 5 + \cos(0) = 6.$$

The solution of the initial-value problem is $y = -\cos x + 6$.



4.52 Solve the initial value problem $\frac{dy}{dx} = 3x^{-2}$, $y(1) = 2$.

Initial-value problems arise in many applications. Next we consider a problem in which a driver applies the brakes in a car.

We are interested in how long it takes for the car to stop. Recall that the velocity function $v(t)$ is the derivative of a position function $s(t)$, and the acceleration $a(t)$ is the derivative of the velocity function. In earlier examples in the text, we could calculate the velocity from the position and then compute the acceleration from the velocity. In the next example we work the other way around. Given an acceleration function, we calculate the velocity function. We then use the velocity function to determine the position function.

Example 4.54

Decelerating Car

A car is traveling at the rate of 88 ft/sec (60 mph) when the brakes are applied. The car begins decelerating at a constant rate of 15 ft/sec².

- How many seconds elapse before the car stops?
- How far does the car travel during that time?

Solution

- First we introduce variables for this problem. Let t be the time (in seconds) after the brakes are first applied. Let $a(t)$ be the acceleration of the car (in feet per seconds squared) at time t . Let $v(t)$ be the velocity of the car (in feet per second) at time t . Let $s(t)$ be the car's position (in feet) beyond the point where the brakes are applied at time t .

The car is traveling at a rate of 88 ft/sec. Therefore, the initial velocity is $v(0) = 88$ ft/sec. Since the car is decelerating, the acceleration is

$$a(t) = -15 \text{ ft/s}^2.$$

The acceleration is the derivative of the velocity,

$$v'(t) = -15.$$

Therefore, we have an initial-value problem to solve:

$$v'(t) = -15, v(0) = 88.$$

Integrating, we find that

$$v(t) = -15t + C.$$

Since $v(0) = 88$, $C = 88$. Thus, the velocity function is

$$v(t) = -15t + 88.$$

To find how long it takes for the car to stop, we need to find the time t such that the velocity is zero.

Solving $-15t + 88 = 0$, we obtain $t = \frac{88}{15}$ sec.

- To find how far the car travels during this time, we need to find the position of the car after $\frac{88}{15}$ sec. We know the velocity $v(t)$ is the derivative of the position $s(t)$. Consider the initial position to be $s(0) = 0$. Therefore, we need to solve the initial-value problem

$$s'(t) = -15t + 88, s(0) = 0.$$

Integrating, we have

$$s(t) = -\frac{15}{2}t^2 + 88t + C.$$

Since $s(0) = 0$, the constant is $C = 0$. Therefore, the position function is

$$s(t) = -\frac{15}{2}t^2 + 88t.$$

After $t = \frac{88}{15}$ sec, the position is $s\left(\frac{88}{15}\right) \approx 258.133$ ft.



4.53 Suppose the car is traveling at the rate of 44 ft/sec. How long does it take for the car to stop? How far will the car travel?

4.10 EXERCISES

For the following exercises, show that $F(x)$ are antiderivatives of $f(x)$.

465.

$$F(x) = 5x^3 + 2x^2 + 3x + 1, f(x) = 15x^2 + 4x + 3$$

$$466. F(x) = x^2 + 4x + 1, f(x) = 2x + 4$$

$$467. F(x) = x^2 e^x, f(x) = e^x(x^2 + 2x)$$

$$468. F(x) = \cos x, f(x) = -\sin x$$

$$469. F(x) = e^x, f(x) = e^x$$

For the following exercises, find the antiderivative of the function.

$$470. f(x) = \frac{1}{x^2} + x$$

$$471. f(x) = e^x - 3x^2 + \sin x$$

$$472. f(x) = e^x + 3x - x^2$$

$$473. f(x) = x - 1 + 4\sin(2x)$$

For the following exercises, find the antiderivative $F(x)$ of each function $f(x)$.

$$474. f(x) = 5x^4 + 4x^5$$

$$475. f(x) = x + 12x^2$$

$$476. f(x) = \frac{1}{\sqrt{x}}$$

$$477. f(x) = (\sqrt{x})^3$$

$$478. f(x) = x^{1/3} + (2x)^{1/3}$$

$$479. f(x) = \frac{x^{1/3}}{x^{2/3}}$$

$$480. f(x) = 2\sin(x) + \sin(2x)$$

$$481. f(x) = \sec^2(x) + 1$$

$$482. f(x) = \sin x \cos x$$

$$483. f(x) = \sin^2(x)\cos(x)$$

$$484. f(x) = 0$$

$$485. f(x) = \frac{1}{2}\csc^2(x) + \frac{1}{x^2}$$

$$486. f(x) = \csc x \cot x + 3x$$

$$487. f(x) = 4\csc x \cot x - \sec x \tan x$$

$$488. f(x) = 8\sec x(\sec x - 4\tan x)$$

$$489. f(x) = \frac{1}{2}e^{-4x} + \sin x$$

For the following exercises, evaluate the integral.

$$490. \int (-1)dx$$

$$491. \int \sin x dx$$

$$492. \int (4x + \sqrt{x})dx$$

$$493. \int \frac{3x^2 + 2}{x^2} dx$$

$$494. \int (\sec x \tan x + 4x)dx$$

$$495. \int (4\sqrt{x} + \sqrt[4]{x})dx$$

$$496. \int (x^{-1/3} - x^{2/3})dx$$

$$497. \int \frac{14x^3 + 2x + 1}{x^3} dx$$

$$498. \int (e^x + e^{-x})dx$$

For the following exercises, solve the initial value problem.

$$499. f'(x) = x^{-3}, f(1) = 1$$

$$500. f'(x) = \sqrt{x} + x^2, f(0) = 2$$

$$501. f'(x) = \cos x + \sec^2(x), f\left(\frac{\pi}{4}\right) = 2 + \frac{\sqrt{2}}{2}$$

$$502. f'(x) = x^3 - 8x^2 + 16x + 1, f(0) = 0$$

503. $f'(x) = \frac{2}{x^2} - \frac{x^2}{2}, f(1) = 0$

For the following exercises, find two possible functions f given the second- or third-order derivatives.

504. $f''(x) = x^2 + 2$

505. $f''(x) = e^{-x}$

506. $f''(x) = 1 + x$

507. $f'''(x) = \cos x$

508. $f'''(x) = 8e^{-2x} - \sin x$

509. A car is being driven at a rate of 40 mph when the brakes are applied. The car decelerates at a constant rate of 10 ft/sec². How long before the car stops?

510. In the preceding problem, calculate how far the car travels in the time it takes to stop.

511. You are merging onto the freeway, accelerating at a constant rate of 12 ft/sec². How long does it take you to reach merging speed at 60 mph?

512. Based on the previous problem, how far does the car travel to reach merging speed?

513. A car company wants to ensure its newest model can stop in 8 sec when traveling at 75 mph. If we assume constant deceleration, find the value of deceleration that accomplishes this.

514. A car company wants to ensure its newest model can stop in less than 450 ft when traveling at 60 mph. If we assume constant deceleration, find the value of deceleration that accomplishes this.

For the following exercises, find the antiderivative of the function, assuming $F(0) = 0$.

515. **[T]** $f(x) = x^2 + 2$

516. **[T]** $f(x) = 4x - \sqrt{x}$

517. **[T]** $f(x) = \sin x + 2x$

518. **[T]** $f(x) = e^x$

519. **[T]** $f(x) = \frac{1}{(x+1)^2}$

520. **[T]** $f(x) = e^{-2x} + 3x^2$

For the following exercises, determine whether the statement is true or false. Either prove it is true or find a counterexample if it is false.

521. If $f(x)$ is the antiderivative of $v(x)$, then $2f(x)$ is the antiderivative of $2v(x)$.

522. If $f(x)$ is the antiderivative of $v(x)$, then $f(2x)$ is the antiderivative of $v(2x)$.

523. If $f(x)$ is the antiderivative of $v(x)$, then $f(x) + 1$ is the antiderivative of $v(x) + 1$.

524. If $f(x)$ is the antiderivative of $v(x)$, then $(f(x))^2$ is the antiderivative of $(v(x))^2$.

CHAPTER 4 REVIEW

KEY TERMS

absolute extremum if f has an absolute maximum or absolute minimum at c , we say f has an absolute extremum at c

absolute maximum if $f(c) \geq f(x)$ for all x in the domain of f , we say f has an absolute maximum at c

absolute minimum if $f(c) \leq f(x)$ for all x in the domain of f , we say f has an absolute minimum at c

antiderivative a function F such that $F'(x) = f(x)$ for all x in the domain of f is an antiderivative of f

concave down if f is differentiable over an interval I and f' is decreasing over I , then f is concave down over I

concave up if f is differentiable over an interval I and f' is increasing over I , then f is concave up over I

concavity the upward or downward curve of the graph of a function

concavity test suppose f is twice differentiable over an interval I ; if $f'' > 0$ over I , then f is concave up over I ; if $f'' < 0$ over I , then f is concave down over I

critical point if $f'(c) = 0$ or $f'(c)$ is undefined, we say that c is a critical point of f

differential the differential dx is an independent variable that can be assigned any nonzero real number; the differential dy is defined to be $dy = f'(x)dx$

differential form given a differentiable function $y = f'(x)$, the equation $dy = f'(x)dx$ is the differential form of the derivative of y with respect to x

end behavior the behavior of a function as $x \rightarrow \infty$ and $x \rightarrow -\infty$

extreme value theorem if f is a continuous function over a finite, closed interval, then f has an absolute maximum and an absolute minimum

Fermat's theorem if f has a local extremum at c , then c is a critical point of f

first derivative test let f be a continuous function over an interval I containing a critical point c such that f is differentiable over I except possibly at c ; if f' changes sign from positive to negative as x increases through c , then f has a local maximum at c ; if f' changes sign from negative to positive as x increases through c , then f has a local minimum at c ; if f' does not change sign as x increases through c , then f does not have a local extremum at c

horizontal asymptote if $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$, then $y = L$ is a horizontal asymptote of f

indefinite integral the most general antiderivative of $f(x)$ is the indefinite integral of f ; we use the notation $\int f(x)dx$ to denote the indefinite integral of f

indeterminate forms when evaluating a limit, the forms $\frac{0}{0}$, ∞/∞ , $0 \cdot \infty$, $\infty - \infty$, 0^0 , ∞^0 , and 1^∞ are considered indeterminate because further analysis is required to determine whether the limit exists and, if so, what its value is

infinite limit at infinity a function that becomes arbitrarily large as x becomes large

inflection point if f is continuous at c and f changes concavity at c , the point $(c, f(c))$ is an inflection point of f

initial value problem

a problem that requires finding a function y that satisfies the differential equation $\frac{dy}{dx} = f(x)$ together with the initial condition $y(x_0) = y_0$

iterative process process in which a list of numbers $x_0, x_1, x_2, x_3 \dots$ is generated by starting with a number x_0 and defining $x_n = F(x_{n-1})$ for $n \geq 1$

limit at infinity the limiting value, if it exists, of a function as $x \rightarrow \infty$ or $x \rightarrow -\infty$

linear approximation the linear function $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f at $x = a$

local extremum if f has a local maximum or local minimum at c , we say f has a local extremum at c

local maximum if there exists an interval I such that $f(c) \geq f(x)$ for all $x \in I$, we say f has a local maximum at c

local minimum if there exists an interval I such that $f(c) \leq f(x)$ for all $x \in I$, we say f has a local minimum at c

L'Hôpital's rule if f and g are differentiable functions over an interval a , except possibly at a , and $\lim_{x \rightarrow a} f(x) = 0 = \lim_{x \rightarrow a} g(x)$ or $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ are infinite, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, assuming the limit on the right exists or is ∞ or $-\infty$

mean value theorem if f is continuous over $[a, b]$ and differentiable over (a, b) , then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Newton's method method for approximating roots of $f(x) = 0$; using an initial guess x_0 ; each subsequent approximation is defined by the equation $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$

oblique asymptote the line $y = mx + b$ if $f(x)$ approaches it as $x \rightarrow \infty$ or $x \rightarrow -\infty$

optimization problems problems that are solved by finding the maximum or minimum value of a function

percentage error the relative error expressed as a percentage

propagated error the error that results in a calculated quantity $f(x)$ resulting from a measurement error dx

related rates are rates of change associated with two or more related quantities that are changing over time

relative error given an absolute error Δq for a particular quantity, $\frac{\Delta q}{q}$ is the relative error.

rolle's theorem if f is continuous over $[a, b]$ and differentiable over (a, b) , and if $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$

second derivative test suppose $f'(c) = 0$ and f'' is continuous over an interval containing c ; if $f''(c) > 0$, then f has a local minimum at c ; if $f''(c) < 0$, then f has a local maximum at c ; if $f''(c) = 0$, then the test is inconclusive

tangent line approximation (linearization) since the linear approximation of f at $x = a$ is defined using the equation of the tangent line, the linear approximation of f at $x = a$ is also known as the tangent line approximation to f at $x = a$

KEY EQUATIONS

- **Linear approximation**

$$L(x) = f(a) + f'(a)(x - a)$$

- **A differential**
 $dy = f'(x)dx$.

KEY CONCEPTS

4.1 Related Rates

- To solve a related rates problem, first draw a picture that illustrates the relationship between the two or more related quantities that are changing with respect to time.
- In terms of the quantities, state the information given and the rate to be found.
- Find an equation relating the quantities.
- Use differentiation, applying the chain rule as necessary, to find an equation that relates the rates.
- Be sure not to substitute a variable quantity for one of the variables until after finding an equation relating the rates.

4.2 Linear Approximations and Differentials

- A differentiable function $y = f(x)$ can be approximated at a by the linear function

$$L(x) = f(a) + f'(a)(x - a).$$

- For a function $y = f(x)$, if x changes from a to $a + dx$, then

$$dy = f'(x)dx$$

is an approximation for the change in y . The actual change in y is

$$\Delta y = f(a + dx) - f(a).$$

- A measurement error dx can lead to an error in a calculated quantity $f(x)$. The error in the calculated quantity is known as the *propagated error*. The propagated error can be estimated by

$$dy \approx f'(x)dx.$$

- To estimate the relative error of a particular quantity q , we estimate $\frac{\Delta q}{q}$.

4.3 Maxima and Minima

- A function may have both an absolute maximum and an absolute minimum, have just one absolute extremum, or have no absolute maximum or absolute minimum.
- If a function has a local extremum, the point at which it occurs must be a critical point. However, a function need not have a local extremum at a critical point.
- A continuous function over a closed, bounded interval has an absolute maximum and an absolute minimum. Each extremum occurs at a critical point or an endpoint.

4.4 The Mean Value Theorem

- If f is continuous over $[a, b]$ and differentiable over (a, b) and $f(a) = 0 = f(b)$, then there exists a point $c \in (a, b)$ such that $f'(c) = 0$. This is Rolle's theorem.
- If f is continuous over $[a, b]$ and differentiable over (a, b) , then there exists a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This is the Mean Value Theorem.

- If $f'(x) = 0$ over an interval I , then f is constant over I .
- If two differentiable functions f and g satisfy $f'(x) = g'(x)$ over I , then $f(x) = g(x) + C$ for some constant C .
- If $f'(x) > 0$ over an interval I , then f is increasing over I . If $f'(x) < 0$ over I , then f is decreasing over I .

4.5 Derivatives and the Shape of a Graph

- If c is a critical point of f and $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$, then f has a local maximum at c .
- If c is a critical point of f and $f'(x) < 0$ for $x < c$ and $f'(x) > 0$ for $x > c$, then f has a local minimum at c .
- If $f''(x) > 0$ over an interval I , then f is concave up over I .
- If $f''(x) < 0$ over an interval I , then f is concave down over I .
- If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .
- If $f'(c) = 0$ and $f''(c) = 0$, then evaluate $f'(x)$ at a test point x to the left of c and a test point x to the right of c , to determine whether f has a local extremum at c .

4.6 Limits at Infinity and Asymptotes

- The limit of $f(x)$ is L as $x \rightarrow \infty$ (or as $x \rightarrow -\infty$) if the values $f(x)$ become arbitrarily close to L as x becomes sufficiently large.
- The limit of $f(x)$ is ∞ as $x \rightarrow \infty$ if $f(x)$ becomes arbitrarily large as x becomes sufficiently large. The limit of $f(x)$ is $-\infty$ as $x \rightarrow \infty$ if $f(x) < 0$ and $|f(x)|$ becomes arbitrarily large as x becomes sufficiently large. We can define the limit of $f(x)$ as x approaches $-\infty$ similarly.
- For a polynomial function $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_n \neq 0$, the end behavior is determined by the leading term $a_n x^n$. If $n \neq 0$, $p(x)$ approaches ∞ or $-\infty$ at each end.
- For a rational function $f(x) = \frac{p(x)}{q(x)}$, the end behavior is determined by the relationship between the degree of p and the degree of q . If the degree of p is less than the degree of q , the line $y = 0$ is a horizontal asymptote for f . If the degree of p is equal to the degree of q , then the line $y = \frac{a_n}{b_n}$ is a horizontal asymptote, where a_n and b_n are the leading coefficients of p and q , respectively. If the degree of p is greater than the degree of q , then f approaches ∞ or $-\infty$ at each end.

4.7 Applied Optimization Problems

- To solve an optimization problem, begin by drawing a picture and introducing variables.
- Find an equation relating the variables.
- Find a function of one variable to describe the quantity that is to be minimized or maximized.

- Look for critical points to locate local extrema.

4.8 L'Hôpital's Rule

- L'Hôpital's rule can be used to evaluate the limit of a quotient when the indeterminate form $\frac{0}{0}$ or ∞/∞ arises.
- L'Hôpital's rule can also be applied to other indeterminate forms if they can be rewritten in terms of a limit involving a quotient that has the indeterminate form $\frac{0}{0}$ or ∞/∞ .
- The exponential function e^x grows faster than any power function x^p , $p > 0$.
- The logarithmic function $\ln x$ grows more slowly than any power function x^p , $p > 0$.

4.9 Newton's Method

- Newton's method approximates roots of $f(x) = 0$ by starting with an initial approximation x_0 , then uses tangent lines to the graph of f to create a sequence of approximations $x_1, x_2, x_3, \dots$
- Typically, Newton's method is an efficient method for finding a particular root. In certain cases, Newton's method fails to work because the list of numbers $x_0, x_1, x_2, \dots$ does not approach a finite value or it approaches a value other than the root sought.
- Any process in which a list of numbers $x_0, x_1, x_2, \dots$ is generated by defining an initial number x_0 and defining the subsequent numbers by the equation $x_n = F(x_{n-1})$ for some function F is an iterative process. Newton's method is an example of an iterative process, where the function $F(x) = x - \left[\frac{f(x)}{f'(x)} \right]$ for a given function f .

4.10 Antiderivatives

- If F is an antiderivative of f , then every antiderivative of f is of the form $F(x) + C$ for some constant C .
- Solving the initial-value problem

$$\frac{dy}{dx} = f(x), y(x_0) = y_0$$

requires us first to find the set of antiderivatives of f and then to look for the particular antiderivative that also satisfies the initial condition.

CHAPTER 4 REVIEW EXERCISES

True or False? Justify your answer with a proof or a counterexample. Assume that $f(x)$ is continuous and differentiable unless stated otherwise.

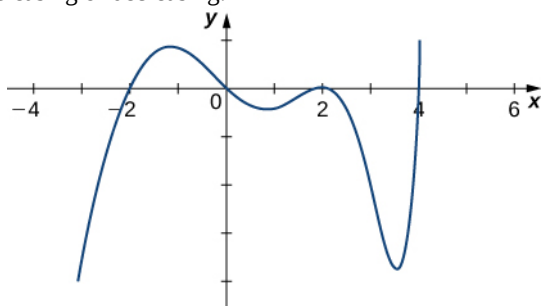
525. If $f(-1) = -6$ and $f(1) = 2$, then there exists at least one point $x \in [-1, 1]$ such that $f'(x) = 4$.

526. If $f'(c) = 0$, there is a maximum or minimum at $x = c$.

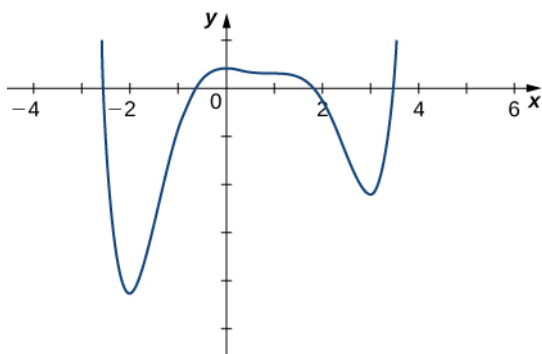
527. There is a function such that $f(x) < 0$, $f'(x) > 0$, and $f''(x) < 0$. (A graphical "proof" is acceptable for this answer.)

528. There is a function such that there is both an inflection point and a critical point for some value $x = a$.

529. Given the graph of f' , determine where f is increasing or decreasing.



530. The graph of f is given below. Draw f' .



531. Find the linear approximation $L(x)$ to $y = x^2 + \tan(\pi x)$ near $x = \frac{1}{4}$.

532. Find the differential of $y = x^2 - 5x - 6$ and evaluate for $x = 2$ with $dx = 0.1$.

Find the critical points and the local and absolute extrema of the following functions on the given interval.

533. $f(x) = x + \sin^2(x)$ over $[0, \pi]$

534. $f(x) = 3x^4 - 4x^3 - 12x^2 + 6$ over $[-3, 3]$

Determine over which intervals the following functions are increasing, decreasing, concave up, and concave down.

535. $x(t) = 3t^4 - 8t^3 - 18t^2$

536. $y = x + \sin(\pi x)$

537. $g(x) = x - \sqrt{x}$

538. $f(\theta) = \sin(3\theta)$

Evaluate the following limits.

539. $\lim_{x \rightarrow \infty} \frac{3x\sqrt{x^2+1}}{\sqrt{x^4-1}}$

540. $\lim_{x \rightarrow \infty} \cos\left(\frac{1}{x}\right)$

541. $\lim_{x \rightarrow 1} \frac{x-1}{\sin(\pi x)}$

542. $\lim_{x \rightarrow \infty} (3x)^{1/x}$

Use Newton's method to find the first two iterations, given the starting point.

543. $y = x^3 + 1$, $x_0 = 0.5$

544. $\frac{1}{x+1} = \frac{1}{2}$, $x_0 = 0$

Find the antiderivatives $F(x)$ of the following functions.

545. $g(x) = \sqrt{x} - \frac{1}{x^2}$

546. $f(x) = 2x + 6\cos x$, $F(\pi) = \pi^2 + 2$

Graph the following functions by hand. Make sure to label the inflection points, critical points, zeros, and asymptotes.

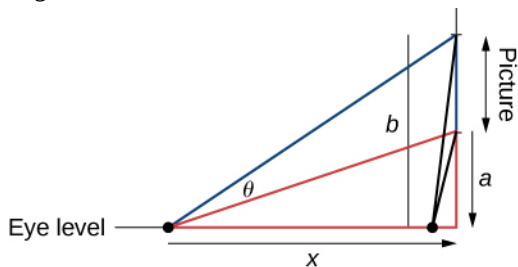
547. $y = \frac{1}{x(x+1)^2}$

548. $y = x - \sqrt{4-x^2}$

549. A car is being compacted into a rectangular solid. The volume is decreasing at a rate of $2 \text{ m}^3/\text{sec}$. The length and width of the compactor are square, but the height is not the same length as the length and width. If the length and width walls move toward each other at a rate of 0.25 m/sec , find the rate at which the height is changing when the length and width are 2 m and the height is 1.5 m .

550. A rocket is launched into space; its kinetic energy is given by $K(t) = \left(\frac{1}{2}\right)m(t)v(t)^2$, where K is the kinetic energy in joules, m is the mass of the rocket in kilograms, and v is the velocity of the rocket in meters/second. Assume the velocity is increasing at a rate of 15 m/sec^2 and the mass is decreasing at a rate of 10 kg/sec because the fuel is being burned. At what rate is the rocket's kinetic energy changing when the mass is 2000 kg and the velocity is 5000 m/sec ? Give your answer in mega-Joules (MJ), which is equivalent to 10^6 J .

551. The famous Regiomontanus' problem for angle maximization was proposed during the 15th century. A painting hangs on a wall with the bottom of the painting a distance a feet above eye level, and the top b feet above eye level. What distance x (in feet) from the wall should the viewer stand to maximize the angle subtended by the painting, θ ?



552. An airline sells tickets from Tokyo to Detroit for \$1200. There are 500 seats available and a typical flight books 350 seats. For every \$10 decrease in price, the airline observes an additional five seats sold. What should the fare be to maximize profit? How many passengers would be onboard?

